Agenda

→ GBDT → M → No of base learners (n-estimators) ⇒ ↑ overfitting

→ depth of learners :→ ↑ overfit

→ ↓ (learning rate)

↑ overfit

$F_m(x) = m_0 + \gamma_1 m_1(x) \ldots\ldots \gamma_2 m_m(x)$

Hyper parameter tuning ⊙ ↑ USE

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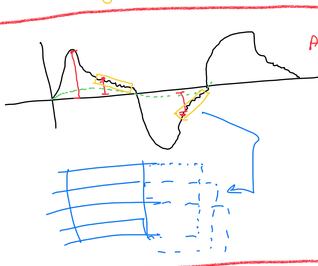
Action 1



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GBDT: Pseudo Residuals + Additive combining

DT

↳ GBDT + Row sampling

↳ stochastic

reduces the variance

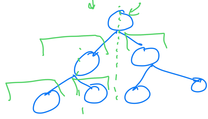
---

XG Boost → Not a new algo

Sk learn → GBDT → B a bad implementation

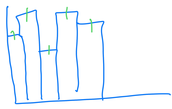
$F_m(x) = m_0(x) + \gamma_1 m_1(x) + \ldots\ldots m_n(x)$

(DT)



GBDT + RS + CS

Numerical Var. / Features



min-split loss ⇒ depth ↓

IG

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Goss

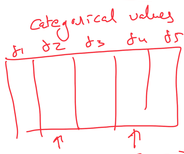
$m^m$ model ⇒ $\{x^{(i)}, r_{m-1}^{(i)}\}_{i=1}^{n}$ → n

d → d'

n → n' ⇒ dropping some point with small error / residual

Graph based models

categorical values → OHE



# features ↓

T ⟷ F → Grouped

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# Stacking → lazy algorithm

↳ kaggle uses this a lot

↳ Real world

Linear model

$w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \ldots\ldots w_d x_d$

$m_0 + \gamma_1 m_1 + \gamma_2 m_2 + \gamma_3 m_3 \ldots\ldots \gamma_n m_n$

↓ no. of base learners

C1 C2 C3 ..... Cn

↓ ↓ ↓ ↓

weights / Prob

High variancy

| T | V |
|---|---|
| 82.8 | 82 |

model → meta model

| T | V |
|---|---|
| 99.9 | 95 |

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# Cascading → errors are expensive

SBT

Best

$f_1$

$f_2$

$f_d$

most that amplify ⇒ Qualityworthy

1 (+)

0 (−)

very complex